

PSTAT 231-Assignment1

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Question 1

Supervised Learning:

Supervised learning uses labeled data, meaning each input X is paired with a known output Y . The goal is to learn a mapping from inputs to outputs.

Unsupervised Learning:

Unsupervised learning uses unlabeled data. The model attempts to discover patterns or structures within the data.

Difference:

The main difference is whether labeled outputs are available. Supervised learning predicts outcomes, while unsupervised learning identifies hidden structures.

Question 2

A regression model predicts continuous numerical values (e.g., house prices, temperature).

A classification model predicts discrete categories (e.g., spam vs. not spam).

Difference:

Regression outputs continuous values, while classification outputs categorical labels.

Question 3

Regression metrics:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Classification metrics:

- Accuracy
- Precision (or Recall / F1-score)

Question 4

Descriptive models:

These models summarize and describe the main features of a dataset without making predictions.

Inferential models:

These models are used to draw conclusions about a population based on sample data, often involving hypothesis testing.

Predictive models:

These models are used to predict future or unknown outcomes based on input data.

Question 5

Mechanistic models:

These are based on known theoretical relationships or domain knowledge and are typically interpretable.

Empirically-driven models:

These rely on data to learn relationships without requiring prior theoretical assumptions.

Differences:

Mechanistic models are theory-driven and interpretable, while empirically-driven models are data-driven and more flexible.

Similarities:

Both aim to model relationships between variables and can be used for prediction.

Interpretability:

Mechanistic models are generally easier to understand because they are based on known principles.

Bias-Variance Tradeoff:

Mechanistic models tend to have higher bias but lower variance. Empirically-driven models tend to have lower bias but higher variance, making them more prone to overfitting.

Question 6

(1) Predicting whether a voter will support a candidate:

This is a **predictive** problem because the goal is to predict an outcome.

(2) Measuring how support changes after personal contact:

This is an **inferential** problem because it aims to understand a causal relationship.

Exploratory Data Analysis

Exercise 1

```
1 ggplot(mpg, aes(x = hwy)) +  
2   geom_histogram(bins = 30, fill = "steelblue", color = "white") +  
3   labs(  
4     title = "Distribution of Highway Miles Per Gallon",  
5     x     = "Highway MPG",  
6     y     = "Count"  
7   ) +  
8   theme_minimal()
```

Listing 1: Histogram of highway MPG

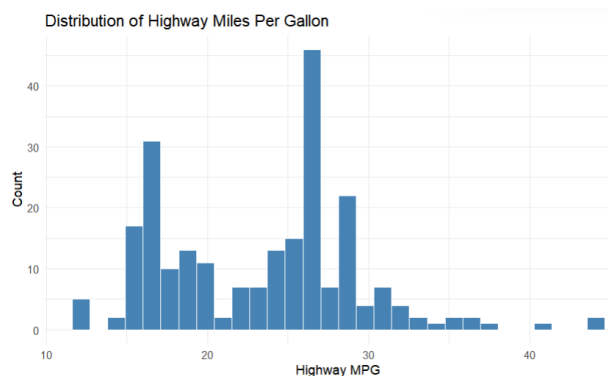


Figure 1: Histogram of highway MPG from the mpg dataset

Observations: The distribution of `hwy` is *right-skewed*: most vehicles cluster between 15 and 30 mpg, with a mode near 26–27 mpg. A secondary peak is visible around 17 mpg, reflecting the large proportion of trucks and SUVs in the dataset. A small number of highly fuel-efficient vehicles (>40 mpg) form a right-hand tail.

Exercise 2

```
1 ggplot(mpg, aes(x = hwy, y = cty)) +
2   geom_point(alpha = 0.5, color = "steelblue") +
3   geom_smooth(method = "lm", se = FALSE, color = "red") +
4   labs(
5     title = "Highway MPG vs. City MPG",
6     x     = "Highway MPG",
7     y     = "City MPG"
8   ) +
9   theme_minimal()
```

Listing 2: Scatterplot of highway vs. city MPG

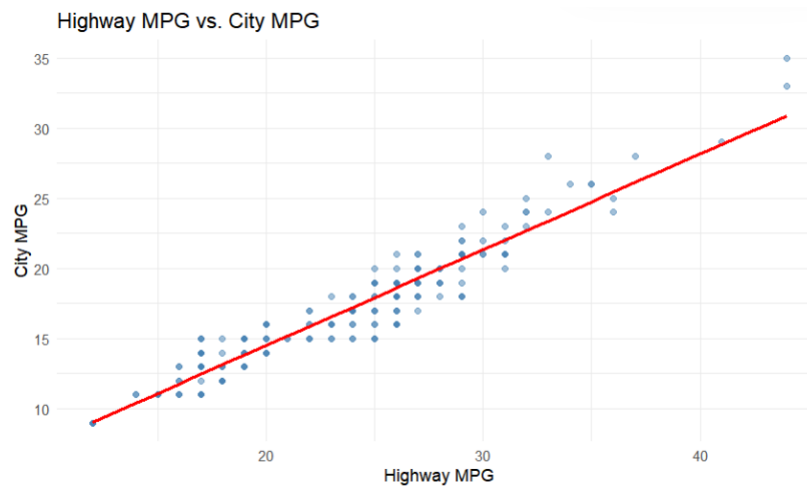


Figure 2: Highway MPG vs City MPG

Observations: There is a *strong positive linear relationship* between `hwy` and `cty`. This is expected: both metrics reflect the same underlying vehicle efficiency (engine displacement, weight, aerodynamics). A vehicle that is fuel-efficient on the highway tends to be fuel-efficient in the city as well. The tight correlation ($r \approx 0.96$) implies that either variable can serve as a reliable proxy for the other.

Exercise 3

```
1 mpg %>%
2   count(manufacturer) %>%
3   ggplot(aes(x = n, y = reorder(manufacturer, n))) +
4   geom_col(fill = "steelblue") +
5   labs(
6     title = "Number of Car Models by Manufacturer",
7     x     = "Count",
8     y     = "Manufacturer"
9   ) +
10  theme_minimal()
```

Listing 3: Bar plot of car counts by manufacturer

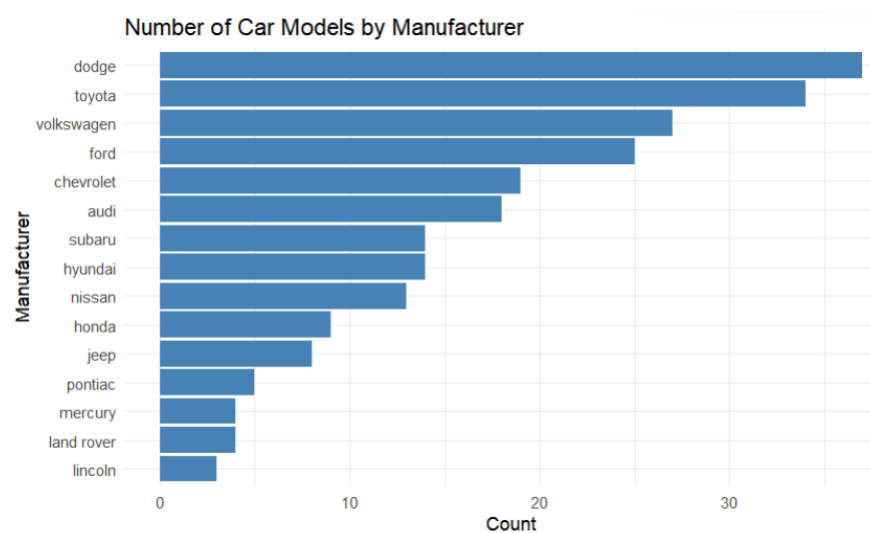


Figure 3: Number of Car Models by Manufactor

Observations: **Dodge** has the largest number of vehicle records in the dataset. **Lincoln** has the fewest (3 vehicles). Ordering the bars by height makes it straightforward to compare manufacturers at a glance.

Exercise 4

```

1 ggplot(mpg, aes(x = factor(cyl), y = hwy)) +
2   geom_boxplot(outlier.shape = NA) +
3   geom_jitter(alpha = 0.3, width = 0.2, color = "steelblue", size = 1.5) +
4   labs(
5     title = "Highway MPG by Number of Cylinders",
6     x     = "Number of Cylinders",
7     y     = "Highway MPG"
8   ) +
9   theme_minimal()

```

Listing 4: Box plot with jitter of highway MPG by cylinder count

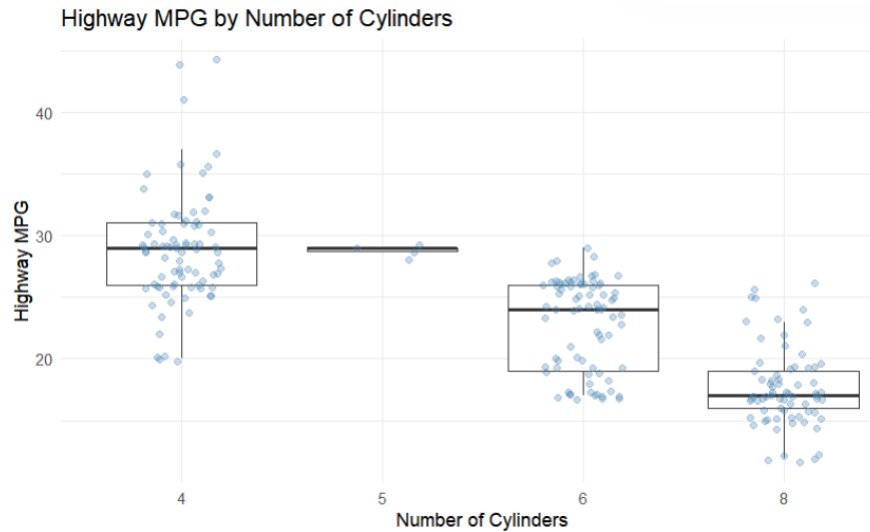


Figure 4: Highway MPG by Number of Cylinders

Observations: There is a clear *negative relationship* between cylinder count and highway MPG: as `cyl` increases from 4 to 8, median `hwy` drops from ≈ 28 to ≈ 17 mpg. Four-cylinder engines are the most fuel-efficient; eight-cylinder engines (typical of trucks and performance cars) are the least. Five-cylinder vehicles are rare in this dataset. The `geom_jitter` layer reveals the density of observations within each group, confirming that 4- and 6-cylinder vehicles are the most common.

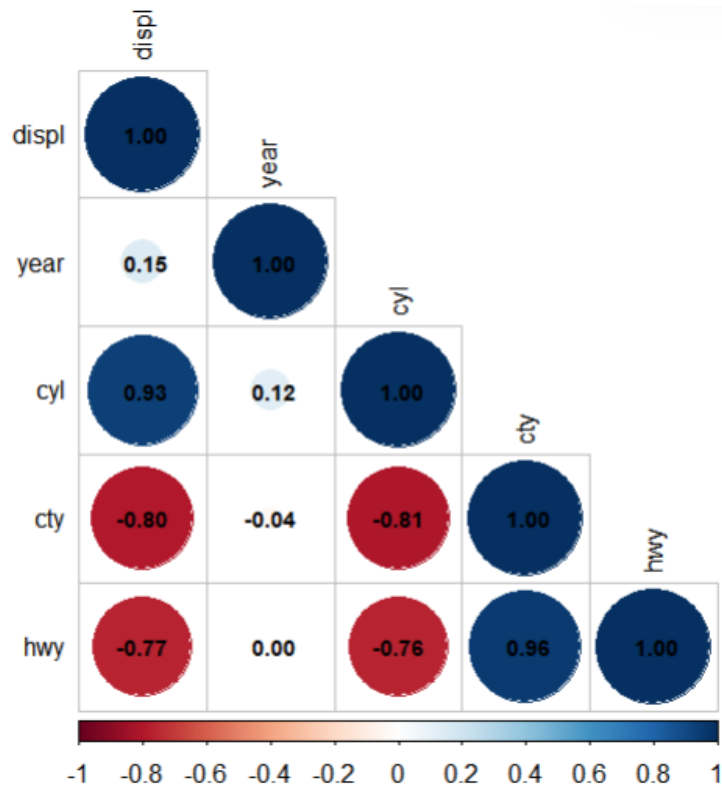
Exercise 5

```

1 mpg_numeric <- mpg %>%
2   select(displ, year, cyl, cty, hwy)
3
4 cor_matrix <- cor(mpg_numeric)
5
6 corrplot(
7   cor_matrix,
8   method = "circle",
9   type    = "lower",
10  tl.cex  = 0.8,
11  tl.col  = "black",
12  addCoef.col = "black",
13  number.cex = 0.7
14 )

```

Listing 5: Lower-triangle correlation matrix for numeric variables



Observations:

- **Strong positive correlations:**

- $\text{cty} \leftrightarrow \text{hwy}$ ($r \approx 0.96$): city and highway fuel economy move almost perfectly together.
- $\text{displ} \leftrightarrow \text{cyl}$ ($r \approx 0.93$): larger engines have more cylinders.

- **Strong negative correlations:**

- $\text{displ} \leftrightarrow \text{cty} / \text{hwy}$ ($r \approx -0.80$): larger displacement $\Rightarrow$ worse fuel economy.
- $\text{cyl} \leftrightarrow \text{cty} / \text{hwy}$ ($r \approx -0.76$): more cylinders $\Rightarrow$ worse fuel economy.

- **Near-zero correlations:** `year` has very weak correlations with all other variables, suggesting that within this dataset's time span, model year is not a strong determinant of fuel efficiency—which is somewhat surprising given improvements in engine technology over time.

All non-trivial relationships are physically intuitive and mutually consistent.

Exercise 6

```

1 class_order <- c("2seater", "compact", "midsize", "minivan",
2                 "pickup", "subcompact", "suv")
3
4 mpg %>%
5   mutate(class = factor(class, levels = class_order)) %>%
6   ggplot(aes(x = hwy, y = class)) +
7   geom_boxplot(outlier.shape = NA, color = "black") +
8   geom_jitter(
9     alpha = 0.35,
10    height = 0.2,

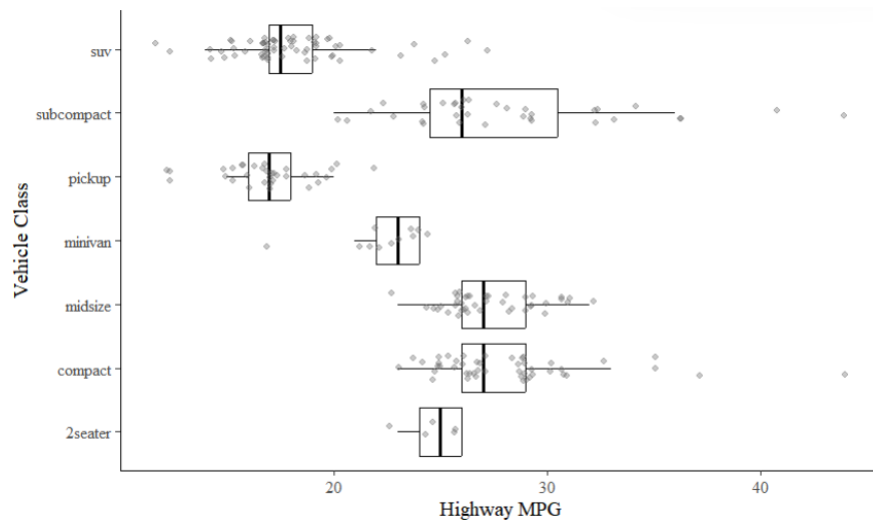
```

```

11     size    = 1.2,
12     color   = "grey40"
13   ) +
14   labs(
15     x = "Highway MPG",
16     y = "Vehicle Class"
17   ) +
18   theme_tufte(base_size = 12) +
19   theme(axis.line = element_line(color = "black"))

```

Listing 6: Tufte-style horizontal box plot with jitter



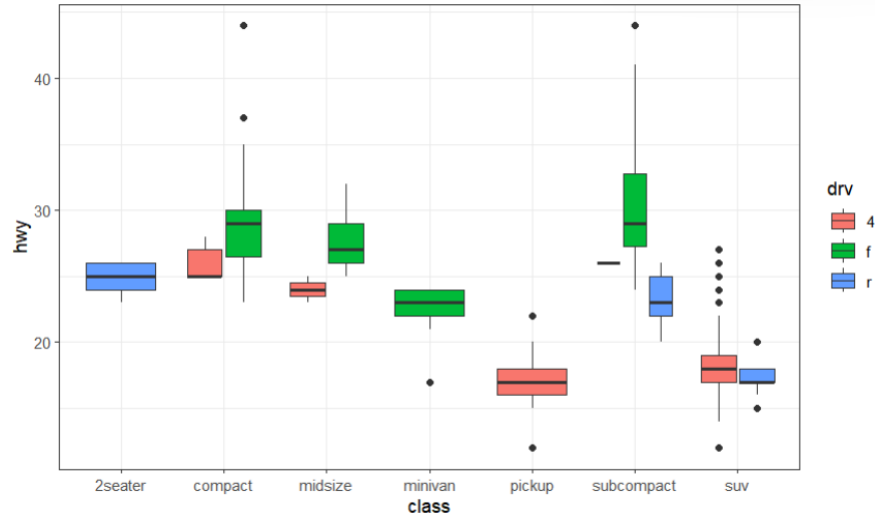
Exercise 7

```

1  ggplot(mpg, aes(x = class, y = hwy, fill = drv)) +
2  geom_boxplot() +
3  scale_fill_manual(
4    values = c("4" = "#F8766D", "f" = "#00BA38", "r" = "#619CFF")
5  ) +
6  labs(
7    x = "class",
8    y = "hwy",
9    fill = "drv"
10 ) +
11 theme_bw() +
12 theme(axis.text.x = element_text(angle = 0, hjust = 0.5))

```

Listing 7: Grouped box plot coloured by drive type



Exercise 7

```

1 ggplot(mpg, aes(x = displ, y = hwy)) +
2   # Points coloured by drive type
3   geom_point(aes(color = drv), size = 2) +
4   # Separate LOESS smoother for each drive type
5   geom_smooth(
6     data      = mpg %>% filter(drv == "4"),
7     method    = "loess",
8     linetype  = "solid",
9     color     = "blue",
10    se        = FALSE
11  ) +
12  geom_smooth(
13    data      = mpg %>% filter(drv == "f"),
14    method    = "loess",
15    linetype  = "dashed",
16    color     = "blue",
17    se        = FALSE
18  ) +
19  geom_smooth(
20    data      = mpg %>% filter(drv == "r"),
21    method    = "loess",
22    linetype  = "dotted",
23    color     = "blue",
24    se        = FALSE
25  ) +
26  scale_color_manual(
27    values = c("4" = "#F8766D", "f" = "#00BA38", "r" = "#619CFF")
28  ) +
29  labs(
30    x      = "displ",
31    y      = "hwy",
32    color  = "drv"
33  ) +
34  theme_bw()

```

Listing 8: Scatter plot of displ vs. hwy with LOESS curves by drive type

